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Batch- 15

HDOC Day-36 — String Operations Using Functions

Algorithm

1. Start.
2. Declare a character array to store the string.
3. Read a string from the user using `fgets( )`.
4. Remove the newline character, if present.
5. Display the menu:
 - Count characters.
 - Print each character on a new line.
 - Count vowels and consonants.
 - Exit.
6. Read the user's choice.
7. Use `switch` to perform the selected operation using a separate function.
8. For **Count Characters**:
 - Initialize `count = 0`.
 - Traverse the string until `'\0'`.
 - Increment `count` for every character.
 - Display the count.
9. For **Print Characters**:
 - Traverse the string until `'\0'`.
 - Print each character on a separate line.
10. For **Vowels and Consonants**:
 - Traverse each character.
 - Check whether it is an alphabet.
 - If it is `a, e, i, o, u` in either case, increment vowel count.
 - Otherwise, increment consonant count.
 - Ignore spaces, digits, and special characters.
11. Repeat the menu until the user selects Exit.
12. Stop.

Test Case Table

Test Case	Input String	Operation	Expected Output
1	Hello	Count characters	5
2	ABC	Print characters	A, B, C on separate lines
3	Education	Count vowels/ consonants	Vowels = 5, Consonants = 4

4	Hello World	Count characters	11 including the space
5	C Programming	Count vowels/ consonants	Vowels = 3, Consonants = 9
6	123 @#	Count vowels/ consonants	Vowels = 0, Consonants = 0
7	Any string	Exit	Program terminates

C Program

```
#include <stdio.h>
```

```
int countCharacters(char str[])
```

```
{    int count = 0;
```

```
    while (str[count] != '\0')
```

```
    {
```

```
        count++;
```

```
    }
```

```
    return count;
```

```
}
```

```
void printCharacters(char str[])
```

```
{
```

```
    int i = 0;
```

```
    printf("\nCharacters in the string:\n");
```

```
    while (str[i] != '\0')
```

```
    {
```

```
        printf("%c\n", str[i]);
```

```
        i++;
```

```
    }
```

```

}

void countVowelsConsonants(char str[])

{
    int i = 0;

    int vowels = 0, consonants = 0;

    char ch;

    while (str[i] != '\0')
    {
        ch = str[i];

        if ((ch >= 'A' && ch <= 'Z') ||
            (ch >= 'a' && ch <= 'z'))
        {
            if (ch == 'A' || ch == 'E' || ch == 'I' ||
                ch == 'O' || ch == 'U' ||
                ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u')
            {
                vowels++;
            }
            else
            {
                consonants++;
            }
        }
    }
}

```

```

        i++;

    }

    printf("Number of vowels = %d\n", vowels);

    printf("Number of consonants = %d\n", consonants);

}

int main()

{

    char str[200];

    int choice;

    int i;

    printf("Enter a string: ");

    fgets(str, sizeof(str), stdin);

    /* Remove newline added by fgets */

    i = 0;

    while (str[i] != '\0')

    {

        if (str[i] == '\n')

        {

            str[i] = '\0';

            break;

        }

        i++;

    }

```

```

do
{
    printf("\n----- MENU ----- \n");

    printf("1. Count characters in the string\n");

    printf("2. Print each character on a new line\n");

    printf("3. Count vowels and consonants\n");

    printf("4. Exit\n");

    printf("Enter your choice: ");

    scanf("%d", &choice);

    switch (choice)
    {

        case 1:

            printf("Number of characters = %d\n",

                countCharacters(str));

            break;

        case 2:

            printCharacters(str);

            break;

        case 3:

            countVowelsConsonants(str);

            break;

        case 4:

            printf("Exiting the program...\n");

```

```

        break;

    default:

        printf("Invalid choice. Please enter 1 to 4.\n");

    }

} while (choice != 4);

return 0;

}

```

Execution

```

[aaryanrajput@aaryans-macbook ~ % nano day36.c
[aaryanrajput@aaryans-macbook ~ % clang day36.c -o day36.c
[aaryanrajput@aaryans-macbook ~ % ./day36.c
Enter a string: hello

----- MENU -----
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Number of characters = 5

----- MENU -----
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2

Characters in the string:
h
e
l
l
o

----- MENU -----
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 2
Number of consonants = 3

----- MENU -----
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting the program...
aaryanrajput@aaryans-macbook ~ % █

```